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Biostatistics III (PPH7095F) - Assignment (2026)

Release date: 12 March

Due date: 2 April (23:00)

Note on submission: Assignments submitted after the due date and time will be penalized by 15% per day late for up to two days. Assignments submitted more than two days late will not be marked. The assignment should be typed and submitted in **PDF format**.

Instructions:

a) There is **no group work** for this assignment. Each student is expected to produce their own independent work. Do not work together on the assignment.
   - If you have questions regarding any aspect of the assignment, please share these during class or make an appointment with the TA/lecturer.

b) Include **a plagiarism declaration** at the beginning of your assignment:

   "I acknowledge that plagiarism is a serious form of academic misconduct and dishonesty. I have read and understood the document, 'Avoiding Plagiarism: A guide for students' from the University of Cape Town. I acknowledge that I understand that the utilisation of open AI tools (such as ChatGPT) without proper acknowledgement and citation is also a form of academic misconduct and is not allowed. I acknowledge and accept the institutional penalties that are applied if I should plagiarise in any submitted work. I have been warned that for coursework assessment: first-time and minor misconduct breaches will result in a mark deduction and warning; and any repeat or major misconduct breaches will result in a failing grade and possible escalation to the Faculty Misconduct Committee (at the course convenor's discretion)".

c) At the beginning of your assignment include AI tools (such as ChatGPT) declaration that either 1) AI tools have not been used or 2) the tools have been used and cite appropriately (**state how they were used**).

d) Put your name and student number on the cover page.

e) **The page limit for the report is 5 pages** (excluding the cover page and appendix).

f) **Round off solutions to 2-3 decimal places, where applicable**.

g) All tables should be formatted (**do not copy-paste or screenshot results from software**).

h) Be concise when summarizing your work. Number your answers as they appear in the assignment.

i) Answer the questions and **submit a PDF file**. Assignment submissions in other formats will be penalized by 5%.

j) Include a **code file added as an appendix to the PDF assignment file**. Failure to include the code file will result in a 10% penalty.

About the assignment: This assignment aims to enhance your skills in performing longitudinal data analysis and accurately interpreting the results. You are free to use any statistical software, but R is highly recommended.

About the data:

The data is derived from a study that examined marital satisfaction levels before a marriage ended. Marital satisfaction was measured on an 11-point scale (0-10) and captured multiple times over up to 400 months of follow-up. The final sample consists of individuals observed in their first marriage across at least three interviews.

Variables:

- *Outcome*: satisfaction level - self-reported marital satisfaction (0-10 scale, 11-point Likert)
- *Time variable*: obs_time - observation time in months from marriage start
- *Baseline covariates*:
    - sex (female or male)
    - age_marriage - age at marriage (years)
    - cohab - premarital cohabitation (1 = yes, 0 = no)
    - income - personal net monthly income (ZAR)
    - hw_all - household workload (1 = all, 0 = some)

Total [57 marks]

Part 0: Data Preparation [not graded but needed to be able to complete the assignment]

Randomly sample 70% of individuals while preserving their full longitudinal profiles. Set a reproducible seed using your birthday + last two digits of your student number in the yyyymmddss format. Assume your longitudinal dataset is called dat.satisfaction and contains one row per observation per person. Below is an example:

library(dplyr)

- # Replace with your birthday + last two digits of your student number in yyyymmddss format
  set.seed(2002112301)
- # Sample 70% of unique person IDs
  id.satisfaction <- sample(unique(dat.satisfaction$id), size = 0.7 *length(unique(dat.satisfaction$id)))
- # Filter dataset to sampled individuals
  dat.satisfaction %>% dplyr::filter(id %in% id.satisfaction) -> dat.satisfaction.new

Use dat.satisfaction.new for all subsequent analyses.

Part 1: Exploratory data analysis [10 marks]

Question 1 [4 marks]

Present and discuss a single table (**Table 1**) summarizing the baseline characteristics of the individuals (first observation per individual). Stratify the table by sex.

Question 2 [3 marks]

Present and interpret a mean longitudinal profile of satisfaction measurements by household workload (**Figure 1**).

Question 3 [3 marks]

Present and interpret a spaghetti plot of satisfaction measurements by premarital cohabitation (**Figure 2**).

Part 2: Generalized estimating equations (GEEs) – [21 marks]

Fit an appropriate GEE to examine the association between marital satisfaction and time, including all covariates (**do not include statistical interactions terms**). Fit two models using different correlation structures:

- GEE 1: First-order autoregressive structure (AR1)
- GEE 2: Unstructured correlation matrix

Question 4 [10 marks]

Present a single table (**Table 2**) with coefficient estimates, standard errors, and 95% confidence intervals (CI) for both GEE models. Use robust (sandwich) standard errors throughout.

Question 5 [2 marks]

Determine the most appropriate correlation structure and justify your choice. State which GEE (GEE 1 or GEE 2) you will use as your final model.

Question 6 [9 marks]

Based on your final GEE model (from Q5), fully interpret the effects of the following predictors on marital satisfaction. For each, provide the estimated coefficient, 95% CI, and an interpretation:

- Time
- Premarital cohabitation
- Household workload

Part 3: Linear mixed effects (LME) model – [20 marks]

Fit an LME model to assess the association between marital satisfaction and time, including all covariates (**do not include statistical interactions terms**). Fit two models with different random-effects structures:

- LME 1: Random intercepts only
- LME 2: Random intercepts and random linear slopes for time

Question 7 [12 marks]

Present a single table (**Table 3**) with fixed-effect coefficient estimates, 95% confidence intervals, and relevant model comparison metrics for both LME models (LME 1 and LME 2). Present only the fixed effects - do not include the random-effects variance components in the table.

Question 8 [2 marks]

Determine the most appropriate random-effects structure and justify your choice. State which LME (LME 1 or LME 2) is your final model.

Question 9 [6 marks]

Compare the estimated effects of the following predictors between your final GEE model (Q5) and your final LME model (Q8):

- Time
- Premarital cohabitation
- Household workload

Part 4: Model validation – [6 marks]

Question 10 [6 marks]

Present two residual diagnostic plots for your final LME model (**Figures 3** and **4**). **One plot should** depict the residuals, **while the other plot should** be related to the random effects. Do these plots look reasonable? Briefly discuss in two to three sentences whether the model assumptions appear to be satisfied.

Software

- Refer to R workshop code